

Tianlong Wang

PhD candidate · Zhejiang University

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Seeking postdoctoral opportunities · Expected PhD completion: June 2027 · Available from July 2027

Research profile

I develop physics-informed and explainable machine-learning methods for debris-flow, landslide and hydrological prediction. My research combines catchment storage and hydrological memory with residual learning, ensemble optimization and uncertainty analysis, with particular attention to small catchments and sparse observations. I seek to extend these methods to related problems in hydrology, geotechnics and engineering monitoring through interdisciplinary collaboration.

Education & international research experience

Zhejiang University · September 2022 – June 2027 (expected)

PhD · Ocean Technology and Engineering · Ocean College

Supervisor: Hongyue Sun

Nanyang Technological University · September 2024 – September 2026

Joint PhD Research Visit (CSC-funded) · School of Civil and Environmental Engineering

Supervisor: Chu Jian

Research competencies

- **Physics-informed prediction:** Water-balance reasoning, hydrological memory and physical consistency constraints for runoff, water-stage and landslide forecasts.
- **Explainable AI and algorithm development:** Ensemble learning, metaheuristic optimization, SHAP analysis and pretrained tabular models for geohazard assessment and sparse-data prediction.
- **Evaluation and uncertainty:** Chronological and spatial evaluation, robust state estimation and sequential prediction intervals.

Selected manuscripts under review

All five: first author. Under review, not accepted; statuses last recorded July–September 2026.

1. **Physics-guided residual stacking for short-lead runoff and stage forecasting in the Fengshuwan catchment, China**
[Journal of Hydrology: Regional Studies](#) · Under review following revision
[Physics-guided residual stacking for short-lead runoff and water-stage prediction](#). Code ↗
2. **Basin Water Storage Degree for Risk-Informed Debris-Flow Warning Support: A Forward-Looking Proof of Concept in a Small Mountain Catchment**
[International Journal of Disaster Risk Reduction](#) · Under review following revision
[A water-balance indicator of antecedent catchment storage for warning-support research](#). Code ↗
3. **Hydrological readiness forecasting for debris flow early warning in a mountain catchment with sparse event records**
[Bulletin of Engineering Geology and the Environment](#) · Under review
[Forecasting warning-relevant hydrological states when confirmed debris-flow records are sparse](#). Code ↗
4. **Robust state estimation and hydrology-informed residual learning for landslide velocity forecasting with prequential prediction intervals**
[Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards](#) · Under review
[Robust motion-state estimation and hydrology-informed residual learning with sequential prediction intervals](#). Code ↗
5. **Sediment-source information improves small-sample prediction of maximum recorded debris-flow volume with a pretrained tabular model**
[Stochastic Environmental Research and Risk Assessment](#) · Under review following revision
[Combining sediment-source information and a pretrained tabular model for small-sample volume prediction](#). Code ↗

Selected published articles

Selected first-author or corresponding-author work. Bold indicates my name; * denotes corresponding author. Full publication list.

1. **DOME: A Dynamic Optimization Meta-Ensemble Framework for Debris-Flow Susceptibility Assessment with SHAP-Based Interpretation**
Tianlong Wang*, H. Yang
Natural Hazards, 122, Article 545. · 2026
First and corresponding author · doi: 10.1007/s11069-026-08307-2
2. **Intelligent Prediction of Post-Seismic Debris Flow Source Volumes: A Physics-Informed Generative Deep Learning Framework**
Tianlong Wang, H. Yang, H. Sun*
Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards, 1–29. · 2026
First author · doi: 10.1080/17499518.2026.2651228
3. **An integrated IKOA-CNN-BiGRU-Attention framework with SHAP explainability for high-precision debris flow hazard prediction in the Nujiang river basin, China**
H. Yang, Tianlong Wang*, N. I. Fomin, S. Xiao, L. Liu
PLOS ONE, 20(6), e0326587. · 2025
Second and corresponding author · doi: 10.1371/journal.pone.0326587
4. **A novel method for predicting debris flow hazard: a multi-strategy fusion approach based on the light gradient boosting machine framework**
Tianlong Wang, Q. Ge, T. Ma, et al.
Stochastic Environmental Research and Risk Assessment, 39, 4867–4890. · 2025
First author · doi: 10.1007/s00477-025-02955-9
5. **Prediction and explanation of debris flow velocity based on multi-strategy fusion Stacking ensemble learning model**
Tianlong Wang, K. Zhang, Z. Liu, T. Ma, R. Luo, H. Chen, X. Wang, W. Ge, H. Sun*
Journal of Hydrology, 638, 131347. · 2024
First author · doi: 10.1016/j.jhydrol.2024.131347
6. **Study and verification on an improved comprehensive prediction model of landslide displacement**
Tianlong Wang, R. Luo, T. Ma, et al.
Bulletin of Engineering Geology and the Environment, 83, Article 90. · 2024
First author · doi: 10.1007/s10064-024-03581-5
7. **Indoor Model Test of Negative Pressure Drainage Method for Treatment of Engineering Mud**
Z. Chu, S. Yu, X. Li, X. Lu, X. Wang, R. Luo, Tianlong Wang*
Journal of Changjiang River Scientific Research Institute, 41(9), 114–122, 129. (In Chinese) · 2024
Corresponding author · doi: 10.11988/ckyyb.20230665
8. **Study on wavelet multi-scale analysis and prediction of landslide groundwater**
Tianlong Wang, D. Peng, X. Wang, B. Wu, R. Luo, Z. Chu, H. Sun*
Journal of Hydroinformatics, 26(1), 237–254. · 2024
First author · doi: 10.2166/hydro.2023.299

Technical skills

- **Programming and research software:** Python; MATLAB; data analysis, model implementation and GitHub-based research code sharing.
- **Machine learning and optimization:** Residual and stacking ensembles; metaheuristic optimization; generative learning; pretrained tabular models.
- **Process-informed modelling:** Catchment water balance; hydrological memory; groundwater response; robust motion-state estimation.
- **Interpretation and evaluation:** SHAP analysis; temporal and spatial evaluation; prediction intervals; small-sample benchmarking.
- **Scientific communication:** English-language publications and research collaboration; two-year CSC-supported research visit at NTU.

Research funding & projects

Young Elite Scientists Sponsorship Program Doctoral Student Special Plan

China Association for Science and Technology · **PI** · January 2025 — December 2026

Lead research on prediction methods and model interpretation for small-catchment debris-flow studies.

Project ID: 156-O-170-0000603-2 Funding: RMB 40,000.

Seed Fund Cultivation Project of Ocean College

Zhejiang University · **PI** · June 2025 — June 2027

Develop a doctoral research programme connecting catchment hydrological states with physics-guided prediction.

Grant No. 2025BS004 Funding: RMB 50,000.

Outstanding Doctoral Dissertation Award Cultivation Grant

Zhejiang University · **PI** · 2025–2026

Doctoral research support for integrating hydrological-state analysis, prediction and interpretation.

Grant No. 825011A · RMB 36,000.

Key Program

National Natural Science Foundation of China · **Core team member** · January 2023 — December 2027

Contribute to the project's hydrology-informed geohazard prediction research; related work includes storage-state and short-lead forecasting studies.

Grant No. 42230702 Total project budget: RMB 2,730,000.

Research software

DOME · Repository ↗

Dynamic ensemble selection and optimization for regional debris-flow susceptibility, with SHAP interpretation.

TabPFNV2.5-DebrisFlow · Repository ↗

Small-sample debris-flow volume prediction using a pretrained tabular model and sediment-source information.

RSRL-HU · Repository ↗

One-day-ahead landslide velocity forecasts with hydrology-informed residual learning and sequential intervals.

Hydrological Readiness Forecasting · Repository ↗

Three-day probabilistic forecasts of warning-relevant hydrological states, evaluated chronologically.

BWSD · Repository ↗

Water-balance-based catchment storage indicator and daily hydrological-state classification.

HydroPIML-RS · Repository ↗

Physics-guided residual stacking for short-lead runoff and water-stage prediction.

Academic service

Selected journal reviewing: Journal of Hydrology; Acta Geotechnica; Bulletin of Engineering Geology and the Environment; Stochastic Environmental Research and Risk Assessment; Applied Intelligence; Array; International Journal of Interactive Multimedia and Artificial Intelligence.

Memberships: Chinese Society for Rock Mechanics and Engineering; Geological Society of China; China Civil Engineering Society.

Selected honors & awards

2025, 2021 · National Scholarship

2023 · China Harbour Engineering Company Scholarship

2022, 2021 · Honorable Mention, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling

Updated 4 October 2026 · Academic homepage · * Corresponding author.